

Sri Lanka Institute of Information Technology



Web-Based Driving License Issuing Service Lab - 03

Group - 026-Y2-S1-MLB-WEB1G2-03

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**SE2030 - Software Engineering
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1. Introduction

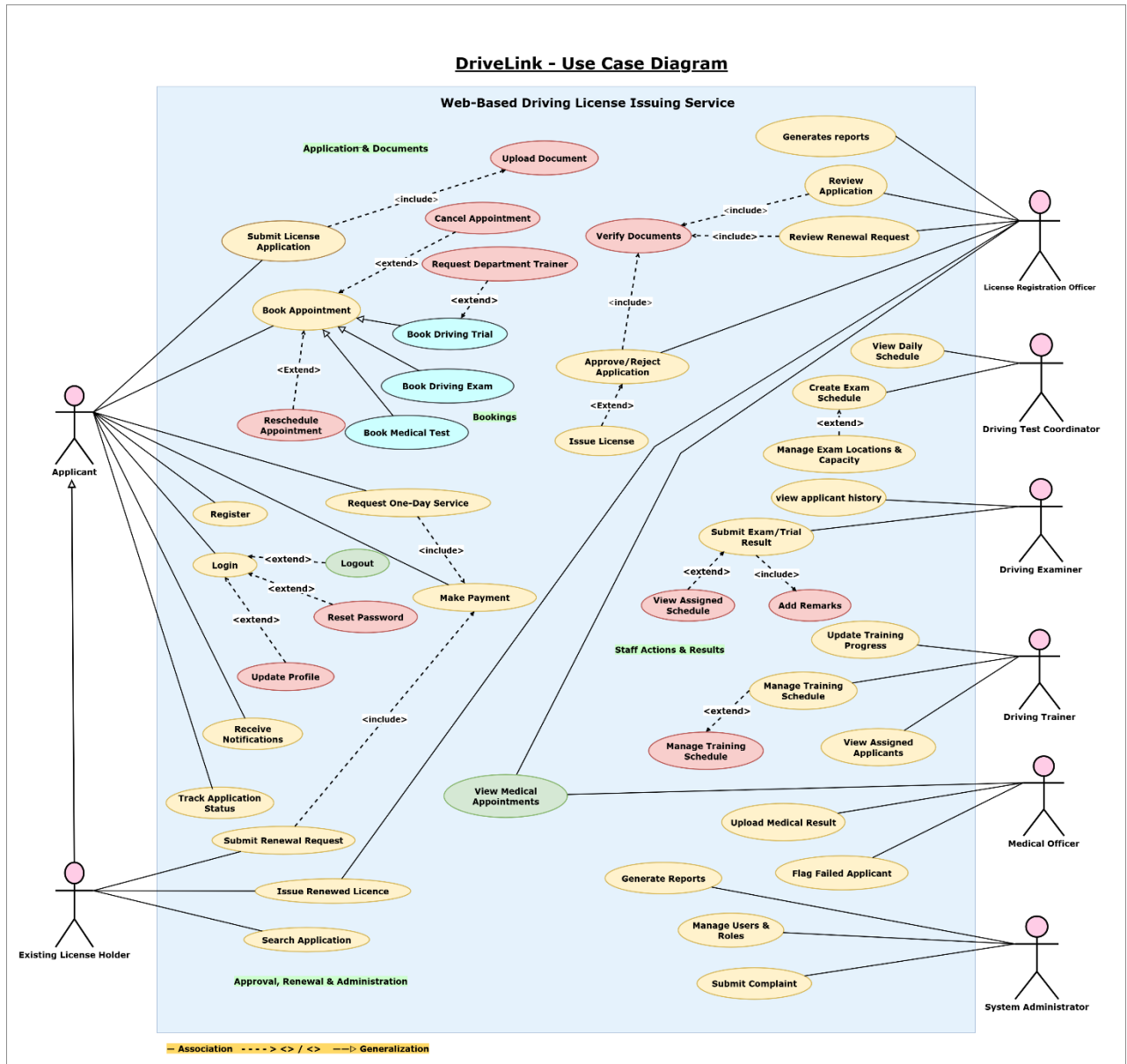
The Web-Based Driving License Issuing Service (DriveLink) is a centralized web platform developed for the Department of Motor Traffic (DMT) that allows citizens to apply for, track, and renew driving licenses online, while enabling officers, examiners, trainers, and administrators to manage the process through dedicated dashboards. This document presents the Use Case Diagram for the system, identifying all actors, their interactions with the system, and the relationships between the various use cases.

2. Actor Identification

The following actors were identified for the DriveLink system. Note: this list was refined from the initial project proposal to include Medical Officer and Driving Trainer as distinct actors, since these roles directly perform key actions within the Medical Test Booking and Driving Trial Booking functions respectively, which were not explicitly separated in the original stakeholder list.

Actor	Type	Description
Applicant	Primary	A citizen who applies for a new driving license
Existing License Holder	Specialized (generalizes from Applicant)	A citizen who already holds a license and wishes to renew it
License Registration Officer	Primary	Reviews applications, verifies documents, approves/rejects, issues licenses
Driving Test Coordinator	Primary	Creates and manages driving exam schedules and slots
Driving Examiner	Primary	Conducts driving exams/trials and submits results
Driving Trainer	Primary	Manages training sessions for applicants requesting a department trainer
Medical Officer	Primary	Manages medical test appointments and uploads medical results
System Administrator	Primary	Manages user accounts, roles, and generates system-wide reports

3. Use Case Diagram



Use Case Diagram for the Web-Based Driving License Issuing Service (DriveLink)

The diagram above illustrates all actors, use cases, and the relationships (association, generalization, include, and extend) present in the system, grouped into four functional areas: Application & Documents, Bookings, Staff Actions & Results, and Approval, Renewal & Administration.

4. Summary of Relationships Used

4.1 Generalization

Child	Parent	Level
Existing License Holder	Applicant	Actor
Book Medical Test	Book Appointment	Use Case
Book Driving Exam	Book Appointment	Use Case
Book Driving Trial	Book Appointment	Use Case

4.2 Include Relationships

Use Case (UC)	Base Use Case	Included Use Case	Rationale
UC-02	Submit License Application	Upload Document	Document upload is always required when submitting a license application.
UC-02	Review Application (Officer)	Verify Documents	Officer must verify documents every time an application is reviewed.
UC-05	Approve/Reject Application	Verify Documents	Final approval always requires document verification before issuance.
UC-05 (ext)	Request One-Day Service	Make Payment	One-Day Service issuance always requires an additional payment.
UC-06	Submit Renewal Request	Make Payment	Renewal always requires the applicant to make a renewal payment.
UC-06	Review Renewal Request (Officer)	Verify Documents	Officer must verify uploaded documents for every renewal request.

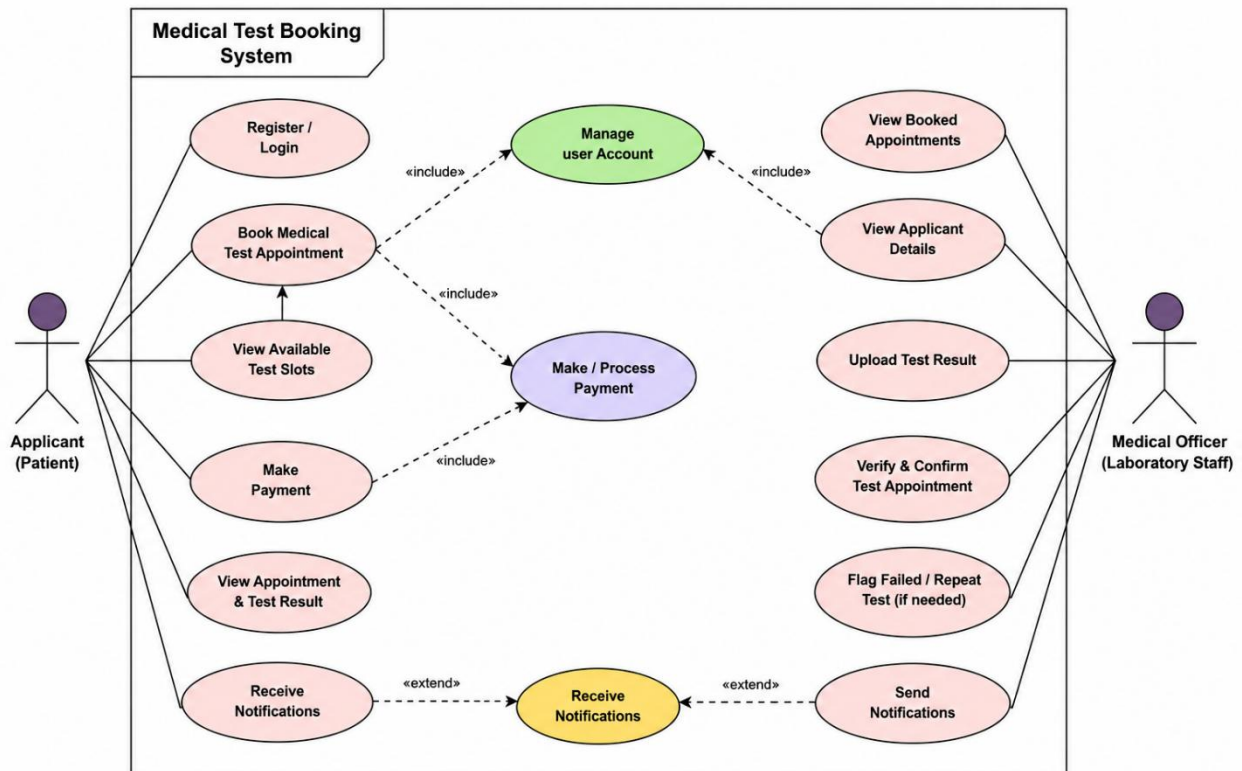
4.3 Extend Relationships

Use Case (UC)	Extending Use Case	Base Use Case	Condition
UC-01 / UC-03 / UC-04	Reschedule Appointment	Book Appointment	Applicant chooses to reschedule an existing booking.
UC-01 / UC-03 / UC-04	Cancel Appointment	Book Appointment	Applicant chooses to cancel an existing booking.
UC-04	Request Department Trainer	Book Driving Trial	Applicant opts for a department-assigned trainer during trial booking.
UC-05	Request One-Day Service	Approve/Reject Application	Applicant has requested expedited (same-day) license issuance with extra payment.

5. Individual Use Case Scenarios

5.1 UC-01: Book Medical Test

Prepared by: Weerasinghe W.P.D.V (IT25100817)

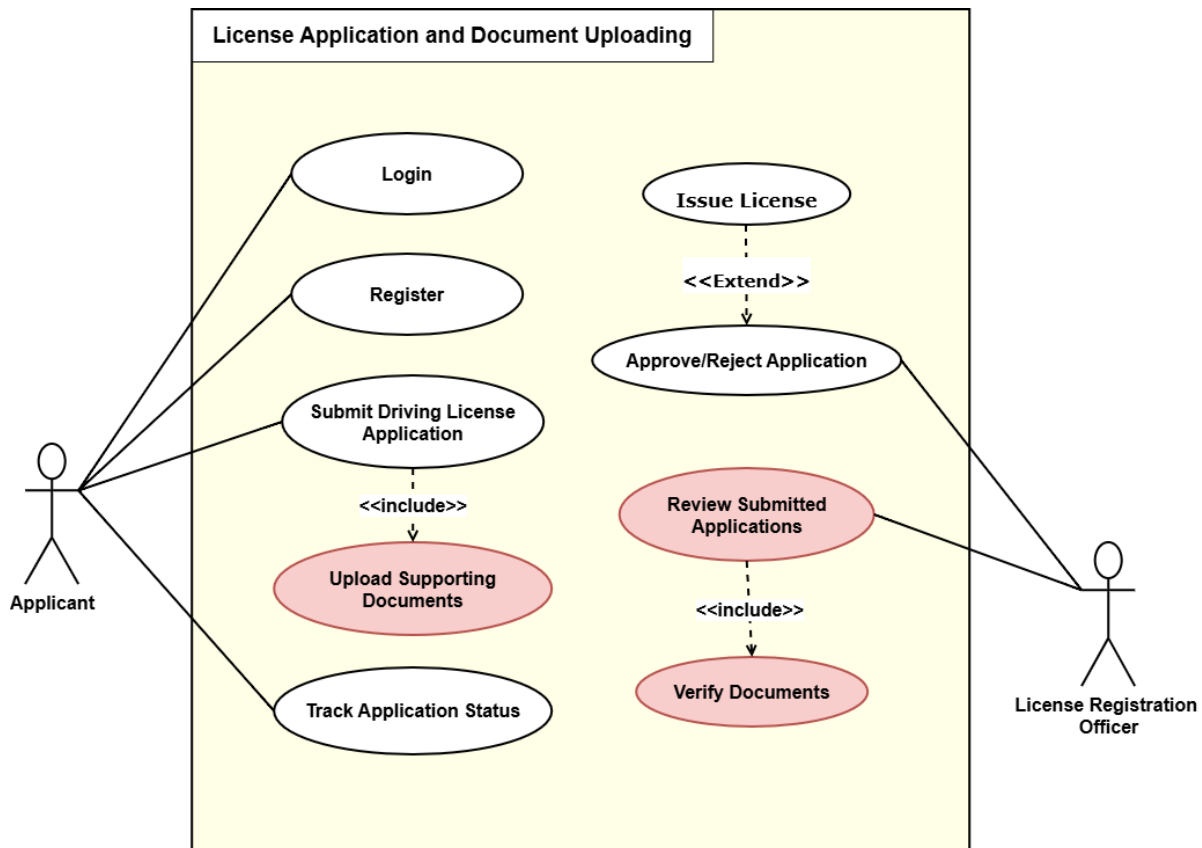


Number	UC-01
Name	Book Medical Test
Summary	Allows an Applicant (Patient) to log in, view available test slots, book a medical test appointment, and make payment. The Medical Officer verifies the appointment, uploads the test result, and notifies the Applicant.
Priority	5 (Highest – required first step in the license application process)
Preconditions	Applicant is registered and logged in to the Medical Test Booking System. Available test slots exist in the system.
Postconditions	A confirmed medical test appointment is recorded in the system. Payment is processed. Applicant receives a booking confirmation and test result notification.
Primary Actor(s)	Applicant

Secondary Actor(s)	Medical Officer																						
Trigger	Applicant logs into the Medical Test Booking System and navigates to the 'Book Medical Test Appointment' section.																						
Main Scenario	<table> <tr> <th>Step</th><th>Action</th></tr> <tr> <td></td><td>Step 1 Applicant logs in; system verifies the session.</td></tr> <tr> <td></td><td>Step 2 Applicant views available dates and times for a medical test.</td></tr> <tr> <td></td><td>Step 3 Applicant selects and confirms a preferred test slot.</td></tr> <tr> <td></td><td>Step 4 Applicant makes the payment; system processes it.</td></tr> <tr> <td></td><td>Step 5 System saves the appointment and assigns a Medical Officer.</td></tr> <tr> <td></td><td>Step 6 Medical Officer logs in; system verifies their access</td></tr> <tr> <td></td><td>Step 7 Medical Officer reviews applicant details and confirms the appointment..</td></tr> <tr> <td></td><td>Step 8 Medical Officer conducts the test and uploads the result.</td></tr> <tr> <td></td><td>Step 9: Applicant views their appointment and test results online.</td></tr> <tr> <td></td><td>Step 10: System sends automated notifications to the applicant for confirmations and results.</td></tr> </table>	Step	Action		Step 1 Applicant logs in; system verifies the session.		Step 2 Applicant views available dates and times for a medical test.		Step 3 Applicant selects and confirms a preferred test slot.		Step 4 Applicant makes the payment; system processes it.		Step 5 System saves the appointment and assigns a Medical Officer.		Step 6 Medical Officer logs in; system verifies their access		Step 7 Medical Officer reviews applicant details and confirms the appointment..		Step 8 Medical Officer conducts the test and uploads the result.		Step 9: Applicant views their appointment and test results online.		Step 10: System sends automated notifications to the applicant for confirmations and results.
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	Step 8a (Test Failed): Medical Officer flags the failed test; system automatically schedules a repeat test.																						
Open Issues	Issue # None identified at this stage.																						

5.2 UC-02: Submit License Application

Prepared by: Simra M.A.F (IT25100815)

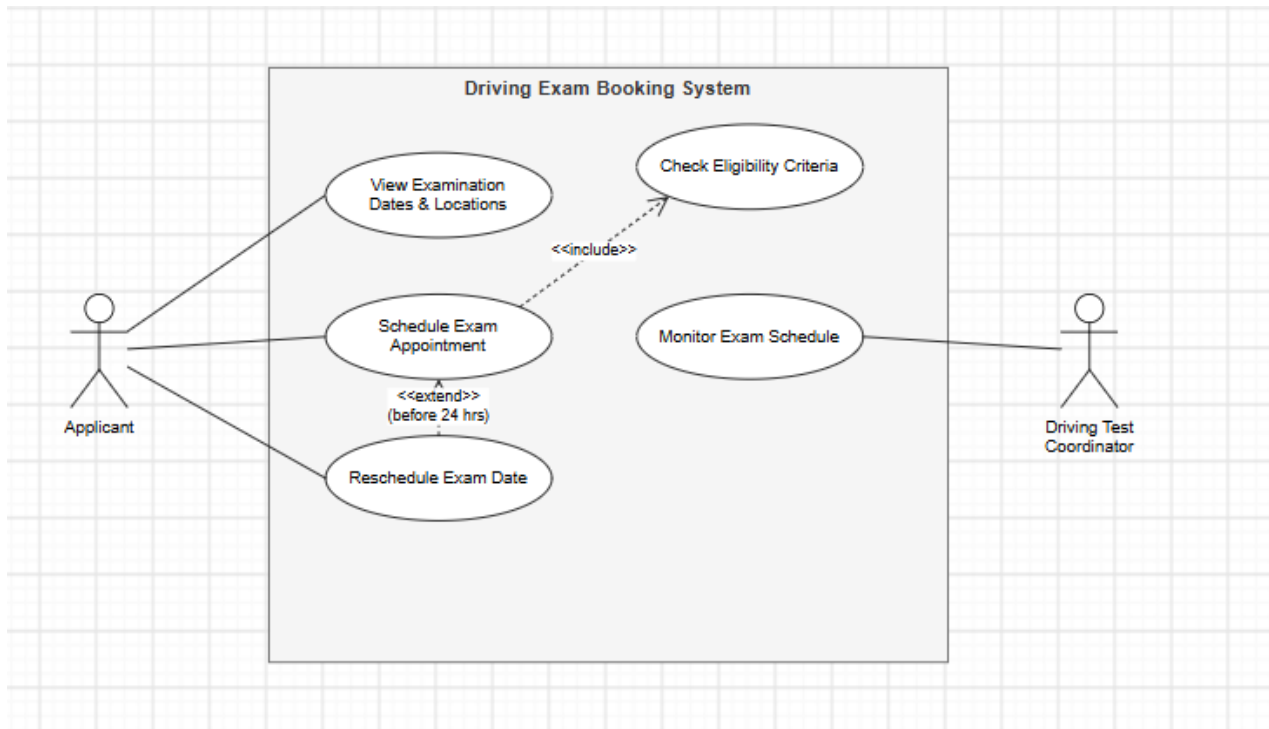


Number	UC-02
Name	Submit License Application
Summary	Allows an applicant to fill in the driving license application form, upload required supporting documents (NIC copy, photograph, medical report), and submit the application for officer review.
Priority	5 (Highest – core function of the system)
Preconditions	Applicant is logged in; medical test has been completed if required.
Postconditions	Application is submitted with 'Pending Review' status and appears in the License Registration Officer's review queue.
Primary Actor(s)	Applicant
Secondary Actor(s)	License Registration Officer
Trigger	Applicant selects 'New Application' from the Citizen Portal dashboard.

Main Scenario	Step Action
	Step 1 Applicant navigates to 'New Application.'
	Step 2 Applicant selects the desired license category and fills in required personal details.
	Step 3 System triggers Upload Document (include) — applicant uploads NIC copy, photograph, and medical report.
	Step 4 System validates file type and size for each uploaded document.
	Step 5 Applicant reviews and submits the completed application.
	Step 6 System sets the application status to 'Pending Review' and notifies the Registration Officer.
	Step 7 Officer opens the application, triggering Verify Documents (include).
	Step 8 Officer approves or rejects the application based on document validity.
Extensions	Step Branching Action
	Step 4a: Invalid file format or size → System rejects the file and prompts the applicant to re-upload.
	Step 8a: Document rejected → System notifies applicant to re-upload the specific document, returning to Step 3.
Open Issues	Issue # None identified at this stage.

5.3 UC-03: Book Driving Exam

Prepared by: Madampage K.S (IT25100851)

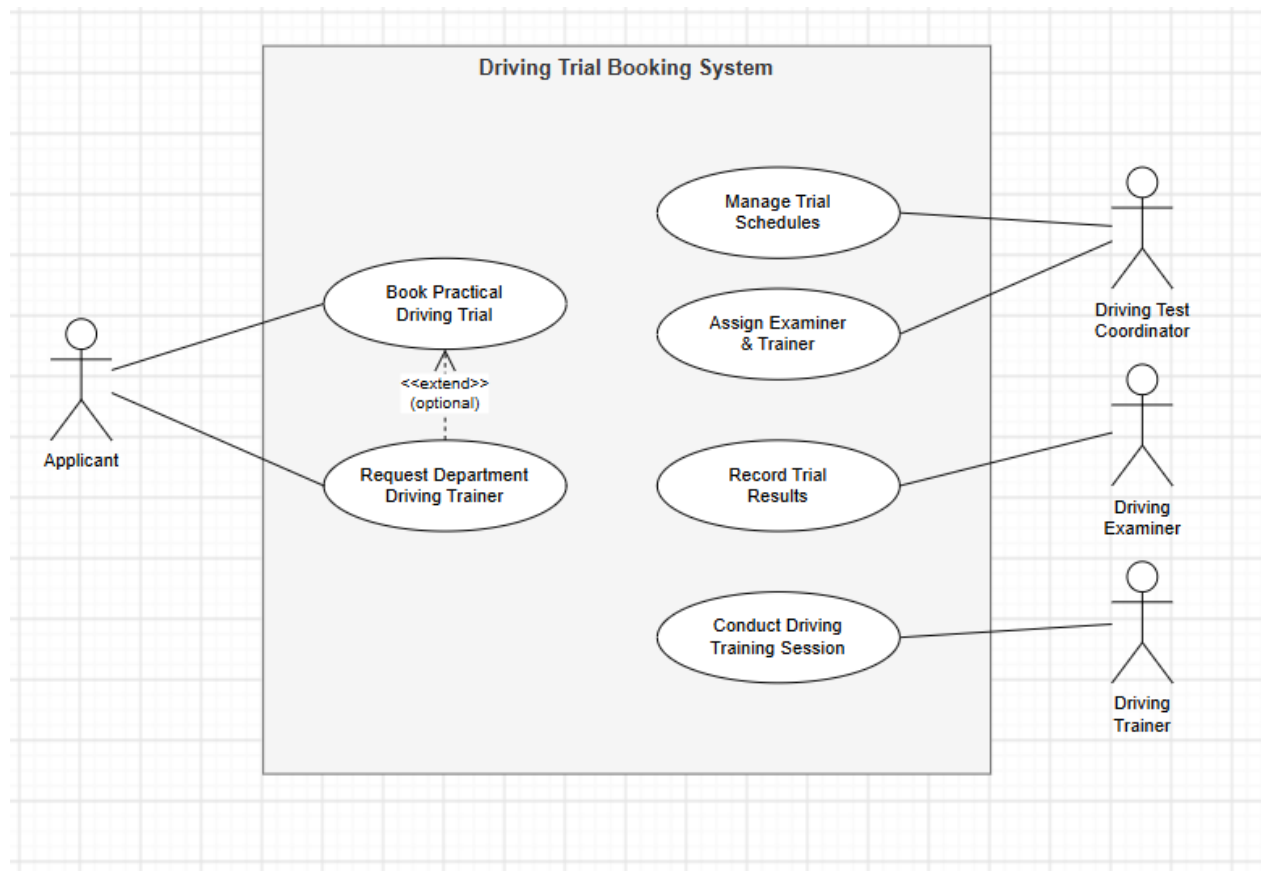


Number	UC-03
Name	Book Driving Exam
Summary	Allows an applicant whose license application has been approved to book a computer-based theory driving examination, selecting a preferred date, time, and location from available slots configured by the Driving Test Coordinator.
Priority	4 (High – required before practical driving trial)
Preconditions	Applicant's license application has been approved to proceed to the exam stage.
Postconditions	Exam appointment is scheduled and assigned to a Driving Examiner.
Primary Actor(s)	Applicant
Secondary Actor(s)	Driving Test Coordinator, Driving Examiner
Trigger	Applicant navigates to 'Book Driving Exam' after receiving application approval notification.

Main Scenario	Step Action
	Step 1 Applicant navigates to 'Book Driving Exam.'
	Step 2 System displays available exam dates and locations, as configured by the coordinator.
	Step 3 Applicant selects a preferred date, time, and location.
	Step 4 System confirms slot availability and creates the booking.
	Step 5 System assigns an available Driving Examiner and notifies the applicant.
	Step 6 Applicant attends the exam on the scheduled date.
	Step 7 Examiner submits the result, and the system updates the applicant's status automatically.
Extensions	Step Branching Action
	Step 4a: Slot unavailable → System prompts the applicant to select a different slot.
	Step 5a: Applicant requests reschedule (more than 24 hours before exam) → Triggers Reschedule Appointment (extend use case).
	Step 5b: Applicant requests cancellation → Triggers Cancel Appointment (extend use case).
Open Issues	Issue # None identified at this stage.

5.4 UC-04: Book Driving Trial

Prepared by: Fernando U.A.E.N (IT25100821)

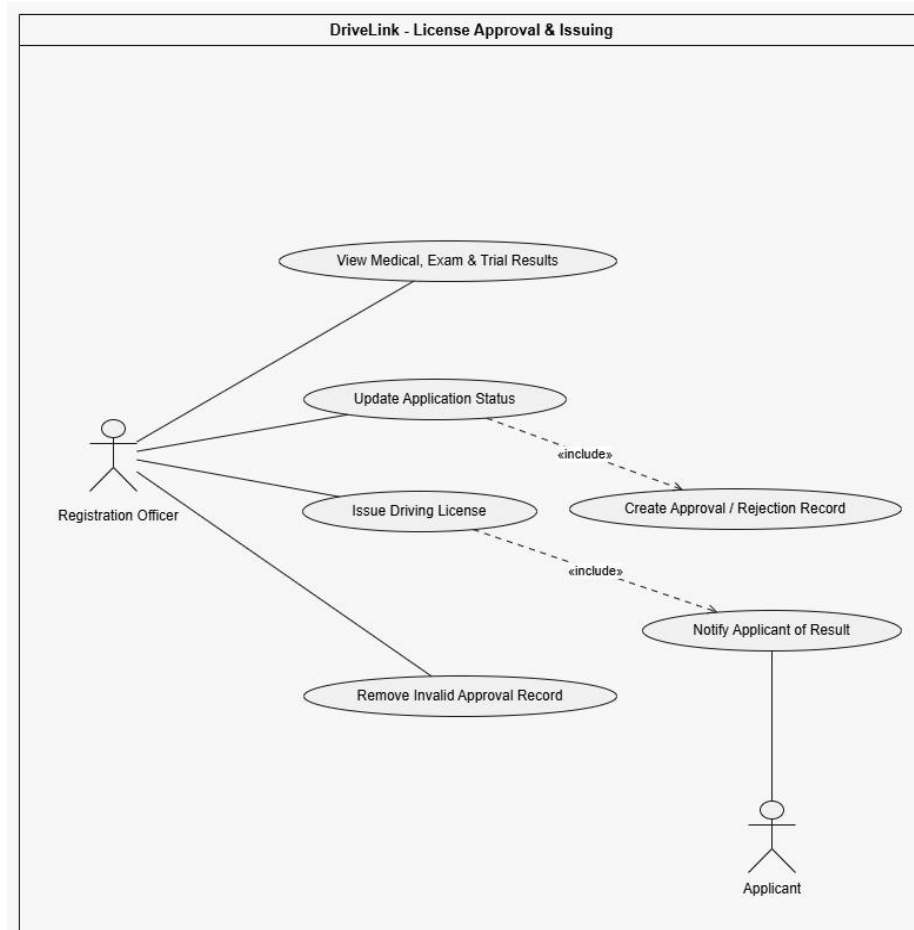


Number	UC-04
Name	Book Driving Trial
Summary	Allows an applicant who has passed the theory exam to book a practical driving trial, optionally request a department-assigned driving trainer, and attend the trial with an assigned Driving Examiner.
Priority	4 (High – final practical step before license approval)
Preconditions	Applicant has passed the driving exam (theory test).
Postconditions	Trial appointment is booked; a trainer is assigned if requested.
Primary Actor(s)	Applicant
Secondary Actor(s)	Driving Test Coordinator, Driving Trainer, Driving Examiner
Trigger	Applicant navigates to 'Book Driving Trial' after receiving theory exam pass notification.

Main Scenario	Step Action
	Step 1 Applicant navigates to 'Book Driving Trial.'
	Step 2 System verifies that the applicant has passed the theory exam.
	Step 3 Applicant selects a preferred trial date and time slot.
	Step 4 System asks whether the applicant requires a department-assigned trainer.
	Step 5 If yes, system triggers Request Department Trainer (extend) and assigns an available trainer.
	Step 6 System confirms the booking and notifies the assigned Trainer and Examiner.
	Step 7 Applicant completes training sessions (if applicable) and attends the trial.
	Step 8 Examiner submits the trial result along with remarks.
Extensions	Step Branching Action
	Step 2a: Theory exam not passed → System blocks booking and displays an eligibility error.
	Step 5a: No trainers available → Request is queued, and the applicant is notified of the delay.
	Step 8a: Applicant fails the trial → Status is set to 'Failed,' and re-booking is allowed after a cooldown period.
Open Issues	Issue # None identified at this stage.

5.5 UC-05: Approve/Reject Application

Prepared by: Thasalogithan S (IT25100830)

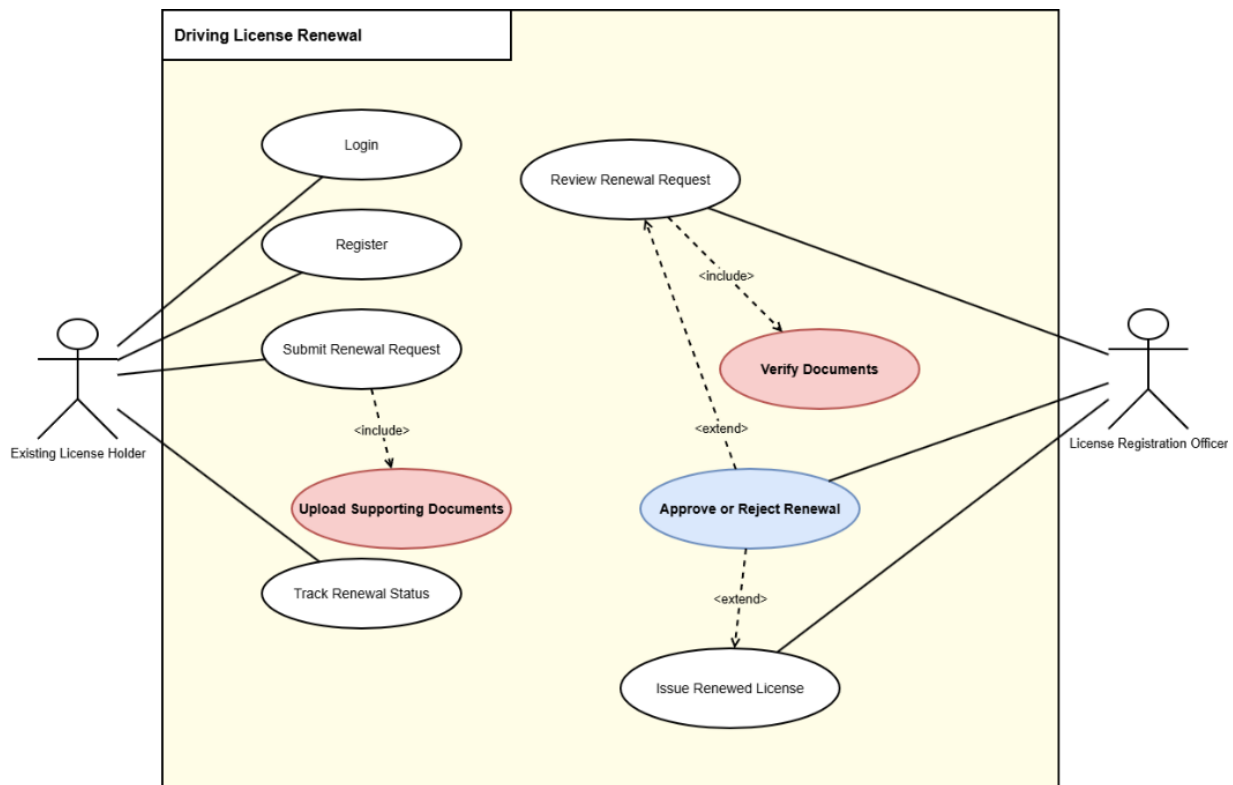


Number	UC-05
Name	Approve/Reject Application
Summary	Enables authorized officers to review applicant eligibility, verify medical, exam, and trial results, update application statuses, create approval/rejection records, issue driving licenses, and manage invalid records within the DriveLink platform.
Priority	5 (Highest – final step in the license issuing workflow)
Preconditions	The Registration Officer is logged into the DriveLink system with authorized privileges. The Applicant has completed all mandatory prerequisites (medical test, written exam, and driving trial).
Postconditions	The applicant's application status is updated (Approved/Rejected).

	• An official approval/rejection record and driving license record are created. • The applicant receives a notification with the result.																		
Primary Actor(s)	License Registration Officer																		
Secondary Actor(s)	Applicant, License Issuing Service Officer																		
Trigger	Registration Officer accesses the DriveLink approval module to process a pending application that has completed medical, exam, and trial stages.																		
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Open Issues	Issue # None identified at this stage.																		

5.6 UC-06: Submit Renewal Request

Prepared by: Karunarathne D.K.K.P (IT25100856)



Number	UC-06
Name	Submit Renewal Request
Summary	Allows an existing license holder whose license is expiring or has expired to submit a renewal request by uploading updated documents and making the renewal payment, for review and approval by the License Registration Officer.
Priority	4 (High – ensures continuous legal compliance for existing drivers)
Preconditions	Applicant holds a driving license that is expiring or has expired.
Postconditions	License is renewed with an updated expiry date, or the renewal request is rejected with a documented reason.
Primary Actor(s)	Existing License Holder
Secondary Actor(s)	License Registration Officer

Trigger	Existing License Holder navigates to 'Renew License' from the Citizen Portal, or receives an expiry notification from the system.																		
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Open Issues	Issue # None identified at this stage.																		

6. Conclusion

This Use Case Diagram identifies all major and minor functions of the DriveLink system, covering eight actors and forty use cases connected through association, generalization, include, and extend relationships. Each of the six group members has defined a comprehensive use case scenario for their assigned major function using the standard Use Case Specification Template, ensuring clear ownership and understanding of the system's core workflows. This model provides the foundation for the subsequent Class Diagram and system implementation phases.